DA1000 — Calculus for Engineers

Tier	A
Phase (order)	1
Do this first	School maths
Estimated effort	10-12 weeks
Unlocks	DA1002, DA3409, DA6360

Roadmap

1. Sequences, limits, series, Taylor and Fourier series.
2. Continuity, Lipschitz, derivatives, gradients, Jacobian, Hessian.
3. Newton's method and gradient descent as applications.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- MIT 18.01 Single Variable Calculus, Fall 2006 (35 videos). MIT 18.01 *Single Variable Calculus* (Fall 2006, David Jerison, official OCW).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- MIT 18.02 Multivariable Calculus, Fall 2007 (35 videos). MIT 18.02 *Multivariable Calculus* (Fall 2007, Denis Auroux, official OCW): partial derivatives, gradients, multiple integrals.
- Mathematics for Machine Learning - Multivariate Calculus (43 videos). Imperial College *Mathematics for ML: Multivariate Calculus*: Jacobians, Hessians, Taylor series, the chain rule for backprop.

⚡ Short videos for intuition and revision

- Essence of calculus (12 videos). 3Blue1Brown *Essence of calculus*: watch first.
- Multivariable calculus (100+ videos). Khan Academy *Multivariable calculus*: short videos for any topic you're stuck on.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–6	Derivatives, limits, continuity, product/quotient/chain rules, implicit differentiation, exp and log. Problem sets from MIT OCW 18.01SC.

Week	Lectures	What to do
2	8–13	Linear/quadratic approximation, curve sketching, max-min, related rates, Newton's method. Code Newton's method in Python.
3	14–18	Mean value theorem, antiderivatives, differential equations, definite integrals, fundamental theorem.
4	19–22	Applications of integration: volumes, work, average value, probability; numerical integration. Code the trapezoid and Simpson rules.
5	24–27	Integration techniques: trig substitution, partial fractions, integration by parts.
6	28–29, 31–32	Parametric curves, arc length, polar coordinates, L'Hospital's rule, improper integrals.
7	33–34	Series, convergence tests, Taylor series. Then multivariable calculus (partial derivatives, gradients) from MIT 18.02 or 3Blue1Brown/Khan (alt).

▼ **All lectures in *MIT 18.01 Single Variable Calculus, Fall 2006* (35 videos)**

1. Lec 1 | MIT 18.01 Single Variable Calculus, Fall 2007
2. Lec 2 | MIT 18.01 Single Variable Calculus, Fall 2007
3. Lec 3 | MIT 18.01 Single Variable Calculus, Fall 2007
4. Lec 4 | MIT 18.01 Single Variable Calculus, Fall 2007
5. Lec 5 | MIT 18.01 Single Variable Calculus, Fall 2007
6. Lec 6 | MIT 18.01 Single Variable Calculus, Fall 2007
7. Lec 7: Exam 1 review | MIT 18.01 Single Variable Calculus, Fall 2007
8. Lec 9 | MIT 18.01 Single Variable Calculus, Fall 2007
9. Lec 10 | MIT 18.01 Single Variable Calculus, Fall 2007
10. Lec 11 | MIT 18.01 Single Variable Calculus, Fall 2007
11. Lec 12 | MIT 18.01 Single Variable Calculus, Fall 2007
12. Lec 13 | MIT 18.01 Single Variable Calculus, Fall 2007
13. Lec 14 | MIT 18.01 Single Variable Calculus, Fall 2007
14. Lec 15 | MIT 18.01 Single Variable Calculus, Fall 2007
15. Lec 16 | MIT 18.01 Single Variable Calculus, Fall 2007
16. Lec 18 | MIT 18.01 Single Variable Calculus, Fall 2007
17. Lec 19 | MIT 18.01 Single Variable Calculus, Fall 2007
18. Lec 20 | MIT 18.01 Single Variable Calculus, Fall 2007
19. Lec 21 | MIT 18.01 Single Variable Calculus, Fall 2007
20. Lec 22 | MIT 18.01 Single Variable Calculus, Fall 2007
21. Lec 23 | MIT 18.01 Single Variable Calculus, Fall 2007
22. Lec 24 | MIT 18.01 Single Variable Calculus, Fall 2007
23. Lec 25 | MIT 18.01 Single Variable Calculus, Fall 2007
24. Lec 27 | MIT 18.01 Single Variable Calculus, Fall 2007
25. Lec 28 | MIT 18.01 Single Variable Calculus, Fall 2007

26. Lec 29 | MIT 18.01 Single Variable Calculus, Fall 2007
27. Lec 30 | MIT 18.01 Single Variable Calculus, Fall 2007
28. Lec 31 | MIT 18.01 Single Variable Calculus, Fall 2007
29. Lec 32 | MIT 18.01 Single Variable Calculus, Fall 2007
30. Lec 33 | MIT 18.01 Single Variable Calculus, Fall 2007
31. Lec 35 | MIT 18.01 Single Variable Calculus, Fall 2007
32. Lec 36 | MIT 18.01 Single Variable Calculus, Fall 2007
33. Lec 37 | MIT 18.01 Single Variable Calculus, Fall 2007
34. Lec 38 | MIT 18.01 Single Variable Calculus, Fall 2007
35. Lec 39 | MIT 18.01 Single Variable Calculus, Fall 2007

Books, notes and websites

- Strang, *Calculus* (OCW Highlights of Calculus)
- 3Blue1Brown *Essence of Calculus*
- Book: Kreyszig, *Advanced Engineering Mathematics*
- Book: Beck, *First-Order Methods in Optimization* (later chapters)

Practice / proof of understanding

Implement Newton's method and gradient descent on three functions and plot the convergence.

What your IITM course sheet says

From the PDF: DA1000

Official description: The goal of this course is to equip engineers with the mathematical tools that they would primarily require for data science and AI. The course will emphasize on understanding and applications of the concepts. On successful completion of the course, participants should be able to: - understand limits, convergence, gradients, and continuity. - solve differential and integral equations. - apply these concepts to analyze iterative algorithms, geometric problems, and optimization. - create models of systems and data using calculus, and analyze their properties.

Official syllabus (trimmed):

> 1. Sequences, limits, convergence theorems and tests. a. Applications to iterative algorithms.
> 2. Series, power series, Taylor series and Fourier series. a. Applications to approximation models.
> 3. Continuity, smoothness, Lipschitz, universal approximation theorems, JL lemma. a. Applications to quantifying neural network robustness.
> 4. Derivatives, subdifferentials, gradients, Jacobian, Hessian. a. Applications to solve optimization problems.
> 5. Root-finding algorithms, Newton’s method, gradient descent and acceleration techniques. a. Applications to train neural networks.
> 6. Evaluating gradients and integrals.

Official textbooks: 1. E. Kreyszig, Advanced Engineering Mathematics, 10th Ed., John Willey & Sons. 2. G. Strang, Calculus, Cambridge press, 3rd ed. 3. A. Beck, First-Order Methods in Optimization, SIAM, 2017. 4. Y. Nesterov, Lectures on Convex Optimization, Springer, 2018.